SET 18

Product Inventory Management

Dataset (CSV)

to understand stock availability and inventory distribution.

Procedure

1. Create the inventory dataset containing Product ID, Product Name, Quantity Available, and Price.
2. Import the dataset into R Studio.
3. Load the required packages such as ggplot2, tidyr, and DT.
4. Create a bar chart to compare available quantities among products.
5. Assign products to categories and generate a stacked bar chart showing inventory distribution.
6. Build an interactive table displaying inventory details.
7. Combine visualizations into a Tableau dashboard for interactive analysis.
8. Examine stock levels to identify products with high and low inventory availability.
9. Interpret inventory patterns for stock management and planning.

R Code

library(ggplot2)

```
library(DT)

inventory <- data.frame(
 ProductID=c(1,2,3),
 ProductName=c("Product A","Product B","Product C"),
 QuantityAvailable=c(250,175,300),
 Price=c(20,15,18)
)

# Q1 Bar Chart
ggplot(inventory,
       aes(ProductName,QuantityAvailable,
           fill=ProductName))+
  geom_bar(stat="identity")+
  labs(title="Inventory Quantity by Product",
       x="Product Name",
       y="Quantity Available")

# Q2 Stacked Bar Chart
inventory$Category <- c("Category A",
                        "Category B",
                        "Category A")

ggplot(inventory,
       aes(Category,
           QuantityAvailable,
```

```
        fill=ProductName))+
  geom_bar(stat="identity")+
  labs(title="Inventory Distribution by Category",
      x="Category",
      y="Quantity")

# Q3 Table
datatable(inventory,
         caption="Inventory Data")

# Q4 Dashboard Components
# Bar Chart + Stacked Bar Chart + Table
```